

## JACC STATE-OF-THE-ART REVIEW

# Contemporary Management of Acute Heart Failure

## From Emergency Presentation to Long-Term Remission

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## ABSTRACT

Acute heart failure (AHF) represents a major global health challenge, with a substantial impact on survival, morbidity, and health care resource utilization. The clinical and economic burden associated with this condition continues to grow, highlighting the need for a timely, structured, and patient-centered approach to care. In light of the expanding body of data and evidence accumulated over recent years, it is necessary to rethink how we approach the recognition and management of AHF. This need has motivated the development of the present review, which aims to provide an updated and practical synthesis of the management of AHF—from early diagnostic assessment and decongestive strategies to the initiation of guideline-directed medical therapy and planning for outpatient transition. The goal is to outline an integrated, holistic, and patient-centered care pathway to facilitate timely recognition and access to guideline-directed medical therapies for AHF. This document is the result of collaboration among an international panel of clinicians with expertise in AHF management. The aim is to offer guidance that is globally applicable across a wide range of clinical settings. Particular attention has been paid to ensuring that the proposed recommendations reflect diverse health care needs while addressing key challenges and threats in the management of patients with AHF. By aligning clinical strategies with patient needs and system capabilities, this review seeks to advance the quality and consistency of AHF care worldwide. (JACC. 2026;■:■-■) © 2026 the American College of Cardiology Foundation. Published by Elsevier. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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ABBREVIATIONS  
AND ACRONYMS

AHF = acute heart failure

ARNI = angiotensin  
receptor-neprilysin inhibitor

BB = beta-blocker

CKD = chronic kidney disease

CRT = cardiac  
resynchronization therapyGDMT = guideline-directed  
medical therapy

HF = heart failure

HFrEF = heart failure with  
reduced ejection fractionICD = implantable  
cardioverter-defibrillatorLVEF = left ventricular  
ejection fractionMRA = mineralocorticoid  
receptor antagonist

NP = natriuretic peptide

NT-proBNP = N-terminal  
pro-B-type natriuretic peptideRASi = renin-angiotensin-  
aldosterone system inhibitorSGLT2i = sodium-glucose  
cotransporter 2 inhibitor

The global incidence of acute heart failure (AHF) remains a major public health concern, with millions of new cases diagnosed annually. AHF affects approximately 1% to 2% of the adult population in developed countries (2-5 per 1,000 person-years) and is increasing in aging and low- and middle-income populations owing to comorbidity burden and limited preventive care.<sup>1-3</sup> Hospitalizations remain frequent and short-term mortality and readmissions continue to challenge outcomes worldwide.<sup>4,5</sup> For all of these reasons, early recognition and rapid initiation of effective therapies are essential.

Recent advances in the diagnosis and management of AHF necessitate an updated review. Accordingly, a globally representative writing committee, including clinicians, nurses, and a patient representative, was convened to summarize current evidence and practice across the continuum from presentation to early after discharge.

In the context of recent European Society of Cardiology (ESC),<sup>6,7</sup> American College of Cardiology/American Heart Association/Heart Failure Society of America guideline updates<sup>8</sup> and American College of Cardiology consensus,<sup>9</sup> this article provides a practical synthesis rather than a position paper. We consider acute hospitalization a disease-modifying inflection point enabling diagnostic clarification, risk stratification, and early initiation of evidence-based therapies influencing long-term outcomes. This review should be used as a clinical framework and bedside guide, and its figures are intended for direct implementation across diverse health care systems worldwide.

## METHODS

Authors were invited to contribute to this review and assigned one or more topics based on their field of expertise. Authors were asked to write a chapter and provide the according literature related to their topics. The process of search and selection of the most relevant literature was limited to articles in English, indexed on Medline, focusing on the advances published during the last 5 years. Each chapter was discussed among the authors during several teleconference sessions, and consensus was achieved on the messages to be included in the article. The draft of the article circulated among the participants to achieve the final form. Literature was refined with a final Medline search in November 2025.

## EARLY HOSPITAL MANAGEMENT OF AHF

**INITIAL EVALUATION AND DIAGNOSIS OF AHF.** AHF is a clinical syndrome characterized by the sudden onset of typical signs and symptoms due to increased filling pressures in the setting of underlying structural or functional cardiac abnormalities.<sup>6,8</sup> This condition may present as a new-onset event or as a worsening of preexisting heart failure (HF). Multiple precipitating factors can lead to fluid accumulation or redistribution, resulting in increased filling pressures and ultimately manifest congestion, which is responsible for the development of symptoms and signs.<sup>10,11</sup> Among them, acute coronary syndrome, arrhythmia, hypertension, infection and inflammation, mechanical causes (such as new-onset valve disease, pulmonary embolism, and pericardial tamponade), and recent noncardiac surgery require specific managements. Clinical severity and manifestations vary widely regardless of left ventricular ejection fraction (LVEF).

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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The initial clinical evaluation of AHF should include assessment of vital signs, symptoms, and clinical signs,<sup>12</sup> and should be supplemented by a comprehensive medical history, focusing on chronic conditions (eg, preexisting heart disease, cardiovascular risk factors) and recent events (eg, medication or dietary changes, anginal symptoms, recent arrhythmia, new-onset valve disease, recent or upcoming noncardiac surgical procedures).<sup>13</sup> This helps to contextualize the current clinical scenario while providing prognostic insights. Given that acute-on-chronic HF is associated with worse outcomes and that certain precipitating factors correlate with higher mortality, this initial assessment is critical to selecting appropriate treatment.<sup>6,14</sup>

Dyspnea, orthopnea, weight gain, fatigue, peripheral edema, abdominal bloating, anorexia, nausea, and nocturia are hallmarks of AHF.<sup>6</sup> Chest discomfort may also be present, making it difficult to differentiate from acute coronary syndrome (ACS), especially because ACS can act as a precipitating factor for AHF and therefore coexist. At this stage, it is critical to rule out ST-segment elevation myocardial infarction (STEMI) or other high-risk ACS and life-threatening arrhythmias by electrocardiography, because these conditions require urgent treatment. Common accompanying signs of HF include peripheral edema, jugular venous distention, a third heart sound, and pulmonary crackles, all of which reflect underlying congestion. In contrast, signs of peripheral hypoperfusion—including cold, clammy skin, altered mental status, and oliguria—suggest cardiogenic shock and warrant immediate recognition and appropriate management. By summarizing these initial steps, the SPLASH acronym provides a structured framework for assessment, improving diagnostic accuracy and enabling timely and appropriate intervention:

- Symptoms
- Past medical history
- Life signs
- Assessment of congestion
- STEMI or equivalent exclusion
- Hypoperfusion

The clinical presentation of AHF often lacks both sensitivity and specificity for a definitive diagnosis, necessitating additional testing. Among the available biomarkers, natriuretic peptides (NPs) play a key role owing to their high sensitivity for detection of underlying cardiac dysfunction. Their measurement on admission in patients with suspected AHF improves diagnostic accuracy beyond clinical assessment alone

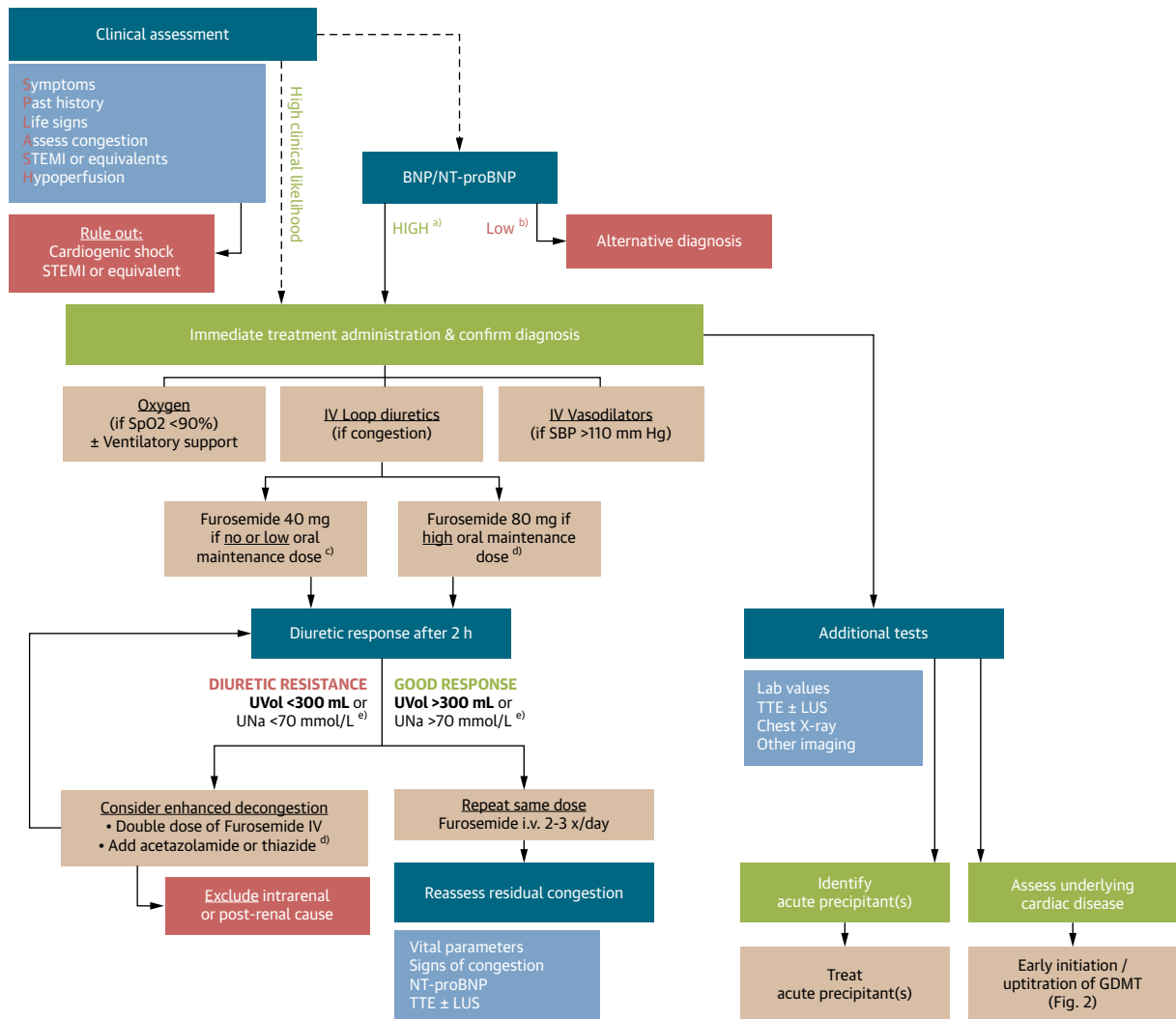
and is therefore strongly recommended, although availability is not yet universal (**Figure 1**).<sup>6,8,15,16</sup>

A low B-type natriuretic peptide (BNP) or N-terminal pro-B-type natriuretic peptide (NT-proBNP) concentration, specifically <100 ng/L for BNP and <300 ng/L for NT-proBNP, effectively rules out AHF with high negative predictive value; this finding prompts the need for an alternative diagnosis.<sup>6</sup> Conversely, a BNP concentration >400 ng/L or an elevated age-adjusted NT-proBNP concentration (>450 ng/L in patients <50 years, >900 ng/L in patients 50-75 years, and >1,800 ng/L in patients >75 years of age) both have higher positive predictive value to identify the diagnosis of AHF.<sup>16</sup>

Some AHF patients may have NPs level at admission that are higher than the rule-out and lower than the rule-in thresholds. This is called the “gray zone” of NPs, where clinical judgment supported by additional diagnostic testing is required to confirm the diagnosis of AHF.<sup>16</sup> Transthoracic echocardiography, pulmonary ultrasound, and other imaging modalities, along with additional laboratory analyses, are required not only to confirm the diagnosis of AHF, but also to identify the underlying cardiac pathology, uncover precipitating factors, assess the severity of congestion, and guide targeted therapeutic strategies (**Figure 1**). Therefore, these tests should also be considered in confirmed cases of AHF.

The measurement of cardiac troponin (cTn) is an important adjunct to the early diagnostic evaluation of AHF; acute coronary ischemia may be a precipitator of the diagnosis and should be evaluated in every case, although caveats to this recommendation exist. When measured among individuals with AHF, concentrations of high-sensitivity (hs) cTn are frequently elevated and often rise or fall in a pattern that might mimic acute myocardial infarction. Mechanistically, troponin release is linked to higher cardiac wall stress, which is associated with worse HF severity; accordingly, higher concentrations of hs-cTn mark individuals at higher risk for worse outcomes in AHF, but they do not a priori reflect acute myocardial infarction in this setting.<sup>17</sup> To make the diagnosis of acute myocardial infarction, clinicians should apply the full criteria from the universal definition for the diagnosis, which necessitates identification of coronary ischemia in the setting of myocardial injury, identification of loss of myocardial function, and evidence for coronary obstruction.<sup>18</sup> Overall, NP concentrations should be interpreted according to age-adjusted thresholds and in light of comorbidities, while hs-cTn values should be interpreted according to dynamic changes and

**FIGURE 1** Early Management of AHF



<sup>a</sup>Rule-in cutoffs: BNP >400 ng/L and NT-proBNP (<50 y: >450 ng/L; 50-75 y: >900 ng/L; >75 y: 1,800 ng/L). Patients presenting with values between rule-out and rule-in thresholds are in the gray zone. Patients with levels in the gray zone need further testing, as reported under "additional tests." <sup>b</sup>Rule-out cutoffs: BNP <100 ng/L and NT-proBNP ≤300 ng/L. <sup>c</sup>Other loop diuretics (eg, bumetanide or torsemide) can be used if locally available. <sup>d</sup>In patients at high risk of diuretic resistance (specifically low eGFR, very high NT-proBNP levels, and very high oral maintenance dose), consider the addition of a second diuretic, such as acetazolamide or a thiazide/thiazide-like diuretic, as locally availability. <sup>e</sup>If available. BNP = B-type natriuretic peptide; eGFR = estimated glomerular filtration rate; GDMT = guideline-directed medical therapy; IV = intravenous; LUS = lung ultrasound; NT-proBNP = N-terminal pro-B-type natriuretic peptide; SBP = systolic blood pressure; SpO<sub>2</sub> = peripheral oxygen saturation; STEMI = ST-segment elevation myocardial infarction; TTE = transthoracic echocardiography; UNa = urinary sodium; UVol = urine volume.

clinical evidence of ischemia rather than absolute values alone.

In addition to cardiac biomarkers, laboratory tests should include a complete blood count, serum electrolytes, and renal and hepatic function. In specific cases, further investigations may be warranted to assess cardiovascular risk factors or to identify underlying conditions, relevant comorbidities, or other

systemic diseases as part of a more comprehensive evaluation during hospitalization.

**EARLY TREATMENT.** The primary goal of early treatment in AHF is the relief of congestion-related symptoms through prompt administration of an appropriate initial dose of diuretic therapy.<sup>19</sup> Because delayed diagnosis is associated with increased in-

hospital mortality, the well established concept of time to treatment has been reinforced.<sup>20,21</sup> **Figure 1** provides a summary of the recommended actions.

On admission, patients should be closely monitored, and initial management should focus on restoring blood pressure (whether initially high or low), arterial oxygenation, ventilation, and treating deleterious arrhythmia. Oxygen supplementation, with or without noninvasive ventilation, depending on the presence of respiratory distress (ie, altered ventilatory mechanics, increased respiratory rate), is recommended when peripheral oxygen saturation (SpO<sub>2</sub>) is <90%.<sup>22</sup>

If the clinical assessment strongly suggests AHF, intravenous (IV) loop diuretics (to bypass impaired enteral absorption due to gastrointestinal congestion) should be started immediately without waiting for confirmatory test results, although completion of the diagnostic workup remains important for subsequent management.

An initial IV administration of 40 mg furosemide (or an equivalent IV dose of another loop diuretic) should be given to oral diuretic-naïve patients or those on low maintenance doses of loop diuretics.<sup>23</sup> For those on maintenance oral higher loop diuretic therapy, an IV bolus of ≥80 mg, or 2 times the oral dose, should be given. Higher loop diuretics can also be considered in patients with chronic kidney disease (CKD) with severely reduced glomerular filtration rate.

Diuretic response should be assessed early to identify patients at risk of diuretic resistance.<sup>23-26</sup> Early addition of a second diuretic can prevent the onset of diuretic resistance.<sup>23</sup> Nonetheless, recent data indicate that the majority of patients receiving an appropriate initial dose of loop diuretics have an adequate diuretic response.<sup>27</sup> Treatment efficacy should be assessed within the first 2 hours after diuretic administration by measuring urine output and, if available, spot urine sodium concentration (**Figure 2**).<sup>28</sup> Urine output ≥300 mL/2 hours or urinary sodium ≥70 to 80 mmol/L are the recommended thresholds for adequate diuretic response.<sup>23,28,29</sup> If these targets are not achieved, the loop diuretic dose should be doubled (up to a maximum furosemide bolus of 250 mg). At this stage, the addition of a second diuretic should again be considered, with further adjustments made until the therapeutic goal is achieved.<sup>28</sup> The maximum recommended daily dose of furosemide is 500 to 600 mg and may be increased to 1,000 mg in patients with impaired kidney function.<sup>21</sup> In addition, it is important to rule out post-renal or intrinsic kidney causes, because these may contribute to the lack of diuretic response and require alternative management strategies. In

cases of refractory diuretic resistance, ultrafiltration should be considered.

Considering that intravenous loop diuretics reach peak efficacy within the first few hours, with sodium excretion returning to baseline by 6 to 8 hours, maintenance of the decongestive effect requires repeated administration up to 3 to 4 times a day or continuous infusion preceded by a bolus until euvoolemia is achieved.

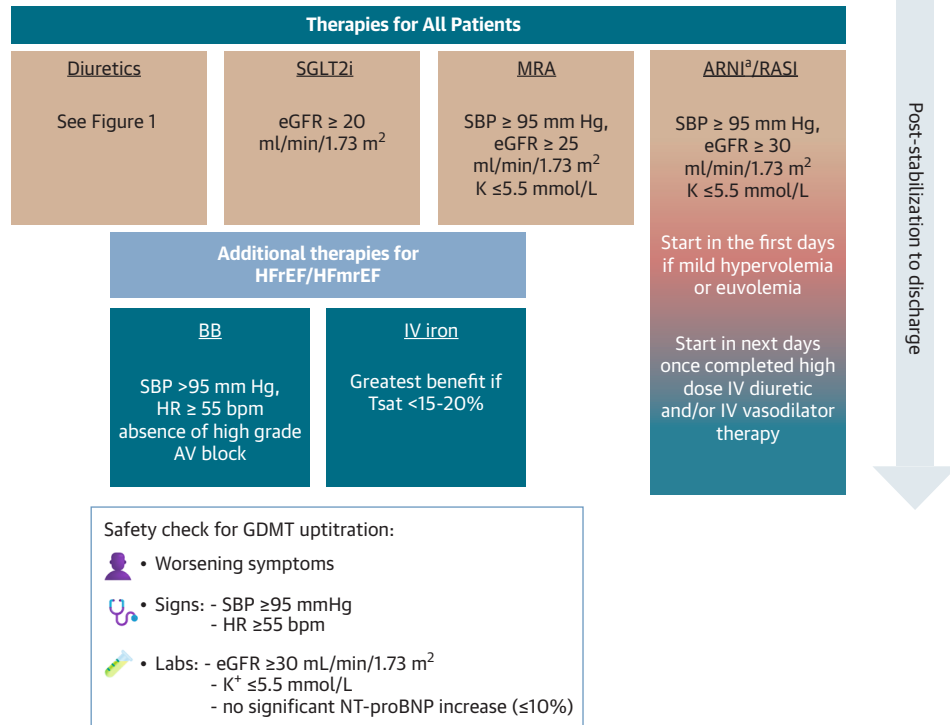
Beyond diuretics, additional initial treatment strategies should be individualized based on the patient's hemodynamic profile.<sup>30</sup> In patients with SBP >110 mm Hg, the addition of intravenous vasodilators, administered according to institutional protocols and adjusted based on patient response, may help to optimize ventricular-arterial coupling, thereby reducing filling pressures and relieving congestion. Among the most frequently used agents are sodium nitroprusside, typically initiated at 0.3 µg/kg per min and uptitrated in increments of 0.3-0.5 µg/kg per min every 10 to 15 minutes to a maximum of 5 µg/kg per min, and nitroglycerine, generally started at 10 to 20 µg/min and increased by 10 to 20 µg/min every 10 minutes up to a maximum dose of 200 µg/min. Decision making in this setting can be supported by the recent scientific statement of the Heart Failure Association of the ESC, which provides a structured framework for the protocolized use of intravenous vasodilators in AHF.<sup>31</sup> Finally, if the patient is already receiving guideline-directed medical therapies (GDMTs), the should be continued in the absence of severe hypotension or marked reduction in kidney function, because they block the mechanisms leading to congestion and improve survival.<sup>6,8,32</sup>

## LONG-TERM DECONGESTION AND INTENSIVE IMPLEMENTATION OF HF THERAPIES: TRICKS AND TIPS

Hospital discharge represents a critical time in AHF care and can affect a patient's subsequent disease course.<sup>12</sup> Indeed, the weeks after hospitalization for AHF are characterized by increased risk of re-admission and death (also called the "vulnerable phase").<sup>33</sup> Three key elements underpin an optimal transition of care:<sup>6,7</sup> to ensure: 1) that patients are fully decongested; 2) that GDMT has been initiated, optimized and prescribed for outpatient use; and 3) that a structured multidisciplinary care program with close follow-up visits is in place.

**DECONGESTION.** Persistent congestion at hospital discharge despite perceived therapy success is common and associated with worse outcomes.<sup>34</sup> There is

**FIGURE 2 Sequencing of HF Therapies in AHF During Hospital Stay: Rapid Initiation in Naïve or Rapid Uptitration of Already Treated Patients**



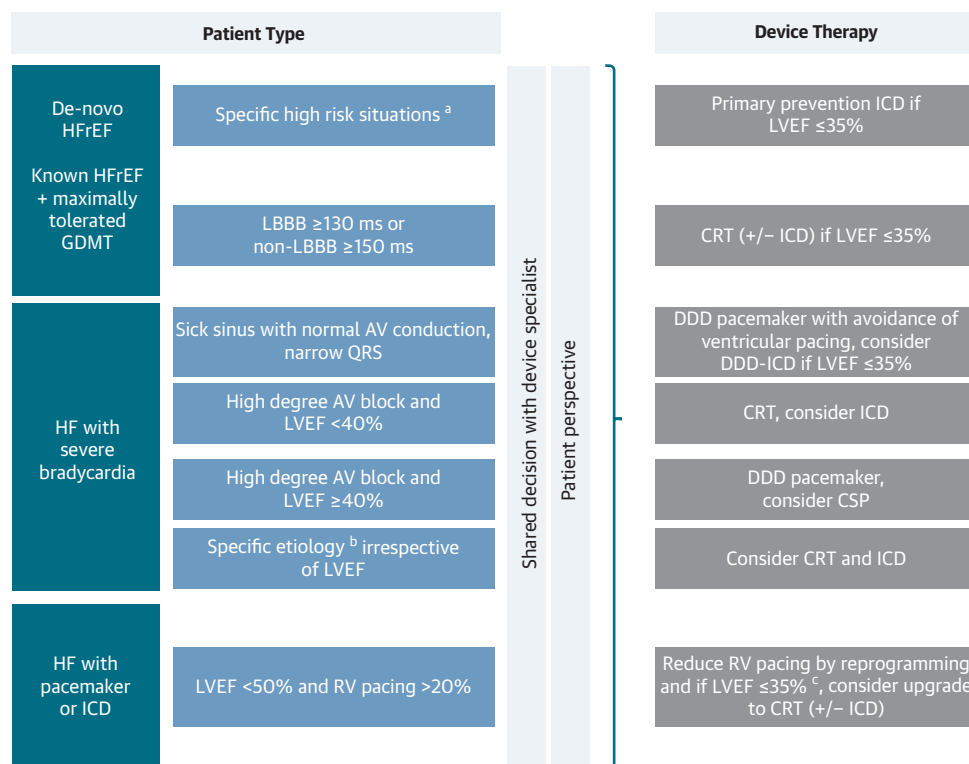
SGLT2is and MRAs should be initiated in the first few days for all patients with HF. In HFrEF and HFpEF, nonsteroidal MRA may be considered if available. In individuals with HFrEF and HFmrEF, ARNIs and BBs should be preferentially added on discharge. If a patient is being moved from an ACEI to an ARNI, a 36-hour washout period is necessary, and consider moving to an ARB during this period. If an ARNI is not feasible owing to availability or cost, then an RASi (ACEI or ARB) should be added. ABB should be added before discharge when congestion is improved, or sooner for management of tachyarrhythmias or ischemia driving HF exacerbation. In individuals with HFpEF, particularly in men with LVEF  $< 60\%$  and women with any LVEF or worsening HF, an ARNI should be added before discharge with ARB as a secondary option if an ARNI is not feasible owing to availability or cost. IV iron should be considered for patients with HFrEF or HFmrEF, with greatest benefit potentially in those with Tsat  $< 15\text{--}20\%$ , based on meta-analyses. Benefit of IV iron is not yet clear in HFpEF.<sup>a</sup> Regardless of eGFR, based on the latest synopsis of the Kidney Disease: Improving Global Outcomes guidelines on chronic kidney disease management,<sup>97</sup> and supported by recent evidence including patients with CKD stage 5,<sup>98,99</sup> bARNI preferred, if available, in women with any LVEF or men with LVEF  $< 60\%$ . ACEI = angiotensin-converting enzyme inhibitor; AHF = acute heart failure; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor-neprilysin inhibitor; AV = atrioventricular; BB = beta-blocker; eGFR = estimated glomerular filtration rate; HF = heart failure; HFmrEF = heart failure with mildly reduced ejection fraction; HFpEF = heart failure with preserved ejection fraction; HFrEF = heart failure with reduced ejection fraction; IV = intravenous; K = potassium; LVEF = left ventricular ejection fraction; MRA = mineralocorticoid receptor antagonist; RASi = renin-angiotensin-aldosterone system inhibitor; SGLT2i = sodium-glucose cotransporter 2 inhibitor; Tsat = transferrin saturation.

no consensus on the best way to assess euvoolemia; in many cases it can be challenging and assessment requires a multimodal approach.<sup>23</sup> A 30% reduction from hospital entry to discharge is considered to be a sign of adequate decongestion.

The transition from IV diuretics to oral diuretics during the index hospitalization for AHF is supported by daily assessment of congestion. In the absence of robust trial data, we consider it reasonable to continue intravenous diuretic therapy until euvoolemia is achieved, then to convert the IV dose to one-

half of its oral equivalent with the aim of maintaining euvoolemia. The discharge dose should be at least equivalent to the pre-admission dose in patients on chronic loop diuretics, or 40 mg of furosemide (or equivalent) in diuretic-naïve patients. Diuretic dose should then be reassessed during early post-discharge follow-up. A subsequent reduction in diuretic dose is generally expected after initiation and up-titration of GDMT.<sup>35,36</sup> No specific class of loop diuretics is preferred at the time of hospital discharge.

**FIGURE 3** Cardiac Implantable Electronic Devices During Index Hospitalization or Soon After



<sup>a</sup>Patients with unexplained syncope; pathogenic genetic variants such as lamin A/C gene, phospholamban, filamin C, or RNA binding motif protein 20; late gadolinium enhancement (on cardiovascular magnetic resonance); inducible sustained monomorphic ventricular tachycardia at programmed electrical stimulation. <sup>b</sup>Cardiomyopathies including sarcoidosis and those related to lamin A/C mutations. <sup>c</sup>If LVEF ≤40% CRT (±ICD) should also be considered (class IIa). BiV = biventricular pacing; CRT = cardiac resynchronization therapy; CSP = conduction-system pacing; DDD = dual-chamber pacing mode; GDMT = guideline-directed medical therapy; HF = heart failure; HFrEF = heart failure with reduced ejection fraction; ICD = implantable cardioverter-defibrillator; LVEF = left ventricular ejection fraction; RV = right ventricular.

**INITIATION, RAPID UPTITRATION OF GDMT FOR HF, AND CLOSE FOLLOW-UP.** Recent studies have shown and ESC 2023 guidelines recommend that early use of the 4 pillars of GDMT is beneficial in all AHF patients regardless of LVEF,<sup>7,37-39</sup> although there are differences in the recommendations of HF therapy between heart failure with reduced ejection fraction (HFrEF) and HF with preserved ejection fraction.

AHF hospitalization is an optimal moment for GDMT initiation and titration to improve patient outcomes and long-term adherence. The STRONG-HF trial showed that an intensive care strategy consisting of predischARGE initiation and rapid uptitration of GDMT, combined with close follow-up, reduced the risk of all-cause death and HF readmission in the 6 months after discharge.<sup>32</sup>

Hospitalized patients benefit from in-hospital initiation or uptitration of 4 major classes of medications, including angiotensin receptor-neprilysin inhibitors (ARNIs; or renin-angiotensin-aldosterone system inhibitors [RASIs] when ARNIs are not available), evidence-based beta-blockers (BBs), mineralocorticoid receptor antagonists (MRAs), and sodium-glucose cotransporter 2 inhibitors (SGLT2i) (Figure 2).<sup>6-9</sup>

Although the ideal sequencing of GDMT is individualized based on the patient's clinical status and laboratory findings, the following expert recommendations provide a general framework based on the mechanisms of these medications. In GDMT-naïve patients with AHF, we recommend early initiation of SGLT2is and MRAs during hospitalization. MRAs help counteract loop diuretic-induced

hypokalemia while potentiating diuresis when combined with loop diuretics.<sup>40</sup> SGLT2is act synergistically with loop diuretics to promote natriuresis and diuresis without triggering neurohormonal activation.<sup>41</sup> In patients admitted with AHF, combination therapy with an MRA and an SGLT2i can be initiated during the index hospitalization, even before LVEF is known.<sup>42-45</sup>

Patients with HFrEF who are already receiving evidence-based BBs (carvedilol, metoprolol succinate, or bisoprolol) therapy before hospitalization may continue if they remain hemodynamically stable and have no evidence for impaired perfusion. In de novo HFrEF, BBs should be initiated after improvement in congestion. If AHF is precipitated by tachyarrhythmia or ischemia, BB therapy may be considered earlier in hospitalization, regardless of LVEF.

ARNIs should be preferred over RASIs in patients with HFrEF, are initiated after initial stabilization with or without vasodilators,<sup>46,47</sup> and are best tolerated in patients with mild hypervolemia or euvolemia. Accordingly, ARNI should be used cautiously in patients actively decongested with high-dose IV diuretics or in hypovolemic patients owing to the potential for resulting hypotension. The PIONEER-HF trial established dosing recommendations based on blood pressure: for systolic blood pressure (SBP) 100 to 119 mm Hg, start ARNIs at 24/26 mg twice daily, and for SBP 120 mm Hg, start ARNIs at 49/51 mg twice daily.<sup>46</sup> When switching a patient from an angiotensin-converting enzyme inhibitor (ACEI) to an ARNI, a 36-hour washout period is required to minimize risk for angioedema, so it is important to consider stopping the ACEI early in the hospital course and switching to an ARB as a short-term bridge.

Additional therapies such as the combination of hydralazine and isosorbide dinitrate may be used in Black patients or in those with contraindications to RASIs or ARNIs; ivabradine and vericiguat may be considered in patients with worsening HF symptoms despite optimal GDMT.

**SPECIAL CONSIDERATIONS.** When implementing GDMT, it is important to remember that most trials were conducted in patients who were hemodynamically stabilized, ie, patients who had not received IV vasoactive medications in the preceding 24 hours and had a systolic blood pressure of  $\geq 100$  mm Hg. As a result, the safety and optimal timing of earlier initiation of HF therapies in more decompensated patients remains unclear, are largely informed by expert consensus rather than trial-based evidence,

and should be approached with caution. Initiating HF drugs in a clinical environment where the dosing requirements for diuretics, inotropes, and vasodilators are rapidly changing can lead to the development of symptomatic hypotension. Once a patient is labeled as “drug intolerant,” clinicians may hesitate to reintroduce such therapies, potentially depriving patients of life-saving treatments that could be safely administered once the patient is chronically stabilized.

Intolerance to GDMT therapy, recurrent hospitalizations, or progressive diuretic resistance with persistent congestion despite escalating doses are key components of the “I NEED HELP” acronym, an established tool for identifying red flags of advanced HF.<sup>48,49</sup> Although primarily consensus based, recognition of these red flags should prompt timely referral to specialized advanced HF centers for comprehensive evaluation and therapeutic planning, including left ventricular assist device implantation and heart transplantation.<sup>49</sup>

Particular attention is required for older and frail patients, who frequently present with a high burden of multimorbidity and greater clinical vulnerability. Evidence guiding GDMT implementation in these populations remains limited, because they are often under-represented in randomized clinical trials, with available data derived from small prospective studies, post hoc analyses, or observational cohorts. Frailty and older age have been consistently identified as factors associated with lower rates of GDMT initiation and uptitration,<sup>50,51</sup> despite the recognition of a high risk for adverse outcomes.<sup>51</sup> Available evidence suggests that advanced age alone should not preclude intensive GDMT.<sup>52,53</sup> A post hoc analyses of STRONG-HF showed that the high-intensity therapy was safe and associated with a reduction in the composite of all-cause mortality or HF readmission across the entire age spectrum, although older patients appeared to derive smaller improvements in patient-reported quality-of-life measures.<sup>54</sup> Taken together, these findings support an individualized cautious but proactive approach to GDMT implementation in older and frail patients, balancing expected benefits against tolerability, comorbidity burden, and patient preferences rather than using age or frailty alone as exclusion criteria.

Finally, special considerations apply to specific cardiomyopathies, such as hypertrophic cardiomyopathy, where BBs are first-line therapy,<sup>55</sup> or cardiac amyloidosis, where data are limited regarding optimal management of AHF and drug intolerances are common.<sup>56</sup>

**OUTPATIENT VISITS.** At the time of hospital discharge, it is critical to ensure that no gaps in aftercare exist and that a clear postdischarge follow-up plan is in place, including scheduled outpatient appointments. Although long-term outpatient optimization extends beyond the acute hospitalization, the early postdischarge period represents a vulnerable phase that plays a pivotal role in shaping subsequent disease trajectory.

For patients discharged after treatment for AHF, early and structured follow-up is critical to monitoring stability and to further optimizing GDMT. Visits should be individualized but are ideally frequent during the first month (weekly to biweekly) and until GDMT at the recommended dose for all HF therapies is fully established,<sup>9</sup> allowing timely evaluation of congestion (combining symptoms, clinical signs, and NT-proBNP level) and reassessment of tolerability by monitoring hemodynamic status (blood pressure and heart rate), renal function, and serum potassium, while facilitating safe and progressive up-titration of therapy. **Figure 2** presents safety checks guiding further up-titration. Based on tolerability and absence of congestion, doses of ARNIs/RASIs, BBs, and MRAs should be increased toward target doses, with a follow-up visit scheduled 1 week later to monitor safety.

There is no rigid algorithm to achieve GDMT at the optimal dose for all HF therapies in a few weeks, so clinical judgment remains essential. For example, if SBP is excessively low and symptomatic, ARNIs/RASIs and MRAs should be maintained at the same dose and not up-titrated on the day of the index visit; however, clinical status often improves within 1 to 2 weeks, allowing up-titration at the subsequent visit. If serum potassium is  $>5.5$  mmol/L, consideration for steps to address hyperkalemia is advised, including use of potassium binders. If estimated glomerular filtration rate is  $<25$  mL/min per  $1.73$  m<sup>2</sup>, use of RASIs (including ARNIs) and MRAs should be reconsidered; if  $<20$  mL/min per  $1.73$  m<sup>2</sup>, SGLT2is should be reconsidered. Use of BBs should be done with caution if resting heart rates are low or if NT-proBNP increases by more than 10% from previous visits. In the latter case, it is preferable to consider increasing the diuretic dose, with reassessment after 1 to 2 weeks, while keeping BBs at the same dose. After this step and clinical improvement, BBs may be up-titrated. In summary, the guiding principle must always be to pursue full implementation of all indicated therapies—within a few weeks and

ideally  $<3$  months—because this is the only proven path to improving outcomes and saving lives.

Regarding diuretic therapy, the goal is to maintain euvolemia with the lowest effective dose, and if at all possible, withdrawal of loop diuretic should be pursued. Safe decongestion is one of the main goals of early hospitalization, and diuretics are the first-line drugs used to achieve this goal. However, by treating only the symptom (fluid overload), diuretics fail to address the cause—the core mechanisms of congestion development—and may lead to adverse compensatory responses.<sup>57,58</sup> In contrast, GDMT (especially neurohormonal blockade) is thought to reduce the internal force that leads to sodium avidity, thereby facilitating effective long-term decongestion even without a direct diuretic effect.<sup>57</sup> In addition, lower sodium avidity should be associated with a lower risk of subsequent decompensation and better diuretic response. In the NATRIUM-HF study, the initiation of ARNI in stable ambulatory patients was associated with improvements in diuretic and natriuretic responses, as well as several adjustments that allow patients to better respond to volume expansion (sodium/volume load).<sup>59</sup> In the STRONG-HF study, assignment to intensive up-titration of neurohormonal blockade was associated with more effective and sustained decongestion by day 90 without an increase in diuretic use;<sup>35</sup> patients with residual congestion at randomization were significantly more likely to be completely decongested by day 90 when using higher doses of GDMT.<sup>35</sup> In recent experience from a rapid outpatient GDMT administration program, diuretic doses were reduced by 50% and nearly one-half of individuals prescribed loop diuretics at baseline were able to stop the agents by program discharge.<sup>60</sup>

Despite the clinical rationale for intensive early follow-up, real-world implementation remains challenging due to health care system constraints, limited specialist availability, organizational barriers, and patient-related factors, such as limited engagement and health literacy. Additional challenges include clinical hesitation and concerns regarding tolerability during early up-titration.

Potential solutions rely on scalable and pragmatic models of care. These include task-sharing approaches supported by targeted education and training programs for cardiologists, internists, and primary care physicians, as well as allied health professionals such as nurses, and pharmacists, allowing GDMT optimization to extend beyond HF specialists alone. The establishment of dedicated GDMT clinics, using a circumscribed allocation of

resources and standardized uptitration protocols, may further improve efficiency. In such models, specialist involvement can be reserved for complex cases, while patients who rapidly achieve target doses can be moved back to routine care.

Equally important is a more active patient-centered approach, with shared therapeutic plans and structured education enabling selected patients to participate in symptom monitoring and, in specific situations, guided self-adjustment of therapy. Finally, digital health tools and hybrid follow-up models combining in-person and remote visits may further support scalable, sustainable, and resource-efficient postdischarge care pathways.

### BENEFITS OF GDMT ON PROGNOSIS AND QUALITY OF LIFE

**EFFECTS ON OUTCOME.** The impact of each of the 4 major HF drug classes on clinical outcomes was first established in chronic HFrEF. Each drug class demonstrated not only a reduced risk of HF hospitalization and cardiovascular death, but also a significant reduction in all-cause mortality. The cumulative relative risk reduction in all-cause mortality with quadruple therapy is estimated to be 73%, resulting in a median survival extension of 7 to 11 years.<sup>61</sup> Subsequently, several GDMT trials were conducted in patients hospitalized for AHF. Although these trials were not powered for clinical outcomes, the exploratory findings on HF-related events are consistent with those in chronic HF, with important lessons learned regarding the impact of GDMT on outcome in those hospitalized with AHF and reassuring safety and tolerability.<sup>62</sup>

In the PIONEER-HF trial, designed to assess the feasibility of early initiation of ARNI, sacubitril/valsartan was associated with a significant 66% lower risk of the composite clinical endpoint of death, rehospitalization for HF, implantation of a left ventricular assist device, or listing for heart transplantation compared with valsartan.<sup>46</sup> In the EMPULSE trial, empagliflozin significantly reduced the incidence of cardiovascular death or HF events by 31% compared with placebo.<sup>63</sup> Similarly, the SOLOIST-WHF trial found that sotagliflozin significantly reduced the incidence of cardiovascular death or HF events by 33% compared with placebo.<sup>64</sup> This is further supported by the STRONG-HF trial, which demonstrated a 34% reduction in death and HF rehospitalization with an intensive treatment strategy of rapid initiation and up-titration of GDMT compared with usual care.<sup>32</sup> In all of these studies, initiation of these key therapies was well tolerated

and without excessive risk; they also reduced hazards compared with delayed postdischarge initiation. Herein lies an important message: Besides facilitating better in-hospital care, initiation of GDMT during hospitalization ensures that individuals with AHF will receive benefits of the therapies after hospitalization. Given that the single best predictor of not receiving GDMT after discharge is absence of GDMT during hospitalization, their initiation is crucial.<sup>65</sup> The approach to the management of AHF has shifted from a singular focus on initial decongestion to the implementation of therapies aimed at achieving long-term improvements in quality of life and overall outcomes.<sup>66</sup>

**HEALTH STATUS AND QUALITY OF LIFE.** The benefits of GDMT implementation extend beyond the aforementioned reductions in mortality and HF hospitalizations. For many patients, improving daily functioning and alleviating symptoms is perceived as more meaningful than simply prolonging life. Accordingly, both health-related quality of life (HRQoL) and overall health status, have gained recognition as clinically relevant, patient-centered endpoints, and have consistently improved with GDMT across a wide range of clinical trials, with the extent of benefit differing according to the subtypes of HF.<sup>67</sup>

For example, in the EMPULSE trial, initiation of empagliflozin during hospitalization for AHF led to early and sustained improvements in HRQoL, with Kansas City Cardiomyopathy Questionnaire scores significantly higher than placebo as early as day 15 and maintained through day 90.<sup>68,69</sup> Moreover, a prespecified subanalysis of the STRONG-HF trial underscored that the early and intensified initiation of GDMT in AHF resulted in significant gains in all dimensions of HRQoL, regardless of patients' baseline self-rated health status.<sup>70</sup>

These findings reinforce the concept that GDMT is not merely life extending, but truly life enhancing. In the context of AHF, prioritizing early and comprehensive optimization of therapy may represent the most effective strategy to improve not only clinical outcomes, but also patients' lived experience.

### COMPREHENSIVE MANAGEMENT: CARDIAC IMPLANTABLE ELECTRONIC DEVICES AND COMORBIDITIES

**COMBINING DRUGS AND CARDIAC IMPLANTABLE ELECTRONIC DEVICES.** Implantation of electronic cardiac devices in HF patients requires close collaboration between device specialists and HF

specialists, who should carefully discuss options with patients before implantation. During hospitalization for AHF, a device implantation is indicated only in cases with severe bradycardia mandating device implantation or cases where secondary preventive implantable cardioverter-defibrillator (ICD) should be considered, ie, in survivors of sudden cardiac arrest or in those with hemodynamically relevant ventricular arrhythmias without a treatable cause. Identifying candidates for later device therapy, however, is already crucial during hospitalization to set the course for follow-up visits.<sup>71,72</sup>

In case of severe bradycardia (eg, sinus node dysfunction) with normal atrioventricular (AV) conduction, a regular dual-chamber pacemaker with minimization of right ventricular (RV) pacing is the first choice. If an individual with LVEF <50% requires ventricular pacing (eg, AV conduction disturbance), biventricular cardiac resynchronization therapy (CRT) or conduction-system pacing (CSP) should be considered regardless of intrinsic QRS duration, because RV pacing is known to be detrimental in HF patients.<sup>73-76</sup> Advanced imaging, including magnetic resonance imaging, could provide additional information to decide whether and which device to implant. If patients with a pacemaker or ICD develop AHF associated with high RV pacing load, consultation with a device specialist is warranted to minimize RV pacing by reprogramming or, if not possible, by device upgrade.<sup>75,76</sup> CRT with biventricular pacing is the first-line option for all patients with LVEF ≤35%,<sup>76</sup> whereas CSP may be the best choice for LVEF >35%.<sup>75</sup> CRT has the potential to restore synchrony, improve LVEF and functional capacity, and reduce HF hospitalization and mortality.<sup>6,76</sup> For this reason, regardless of the presence of bradyarrhythmias, CRT should be considered in HFrEF patients with wide QRS ≥130 ms, especially those with left bundle branch block morphology or QRS ≥150 ms.

Although cardiac implantable electronic devices such as CRTs or ICDs are of benefit in appropriately selected patients with reduced LVEF, it is critical to allow effects of GDMT on cardiac remodeling before proceeding; optimal GDMT deployment may have substantial impact on device eligibility. For example, in the PROVE-HF study, among study participants eligible for primary ICD placement on the basis of a baseline LVEF ≤35%, by 6 months after ARNI initiation 32% were no longer eligible on the basis of low LVEF, and by 12 months 62% were no longer eligible.<sup>77</sup> Those with very low starting LVEF, markedly elevated NT-proBNP at baseline or after ARNI

treatment, and other high risk features were less likely to improve their LVEF to >35%.<sup>78</sup> Recent data from a GDMT clinic showed nearly identical results from PROVE-HF, with 65% of ICD-eligible individuals improving their LVEF to >35%.<sup>60</sup> A suggested strategy for the approach to device eligibility and GDMT administration is presented in **Figure 3**.

There are many situations where proceeding directly to invasive management for rhythm diagnoses is indicated. In patients with atrial tachyarrhythmias precipitating AHF who are considered to be unsuitable for medical therapy or specific ablation, a “pace and ablate” strategy including CSP and AV node ablation should be considered.<sup>75</sup> In addition, individuals with specific HF etiologies predisposing to an increased risk of sudden cardiac death, such as cardiac sarcoidosis or laminopathies, should be considered for early implantation with a device capable of both pacing and ICD.<sup>71,72</sup>

#### HOW TO MANAGE THE MOST FREQUENT COMORBIDITIES.

Patients with AHF often have a high burden of cardiac and noncardiac comorbidities, whose prognostic impact varies according to both the type and the number of comorbid conditions, and are associated with more complex management and increased risk of adverse outcomes.<sup>79</sup> CKD, arrhythmias, valvular heart disease, and iron deficiency are among the most prevalent and treatment-relevant. The prevalence, impact on management, and considerations for their specific management are outlined in **Table 1**.

Individuals with CKD are beset with higher risk for adverse cardiovascular outcomes and limitations regarding GDMT options. Management of worsening kidney function in the context of AHF hospitalization exceeds the scope of the present review, but recent consensus guidance for its management is available.<sup>80</sup> Nonetheless assiduous decongestion and application of GDMT is possible in those with CKD, and continuation of ARNIs or SGLT2is in these patients offers clinical benefits without added safety risks.<sup>81,82</sup>

Because arrhythmia is a frequent cause of HF decompensation, rhythm management is frequently an important component to AHF management, as described above and in **Table 1**.

Valvular heart disease plays an important role in the cause of AHF and in its optimal management. Specifically, aortic and mitral disease may lead to worsened symptoms and volume overload in AHF. A recent review on the management of decompensated valvular heart disease in HF provides useful guidance.<sup>83</sup> Use of transcatheter aortic valve replacement may represent an important option for the care of

**TABLE 1 Special Considerations in Presence of Common Comorbidities**

| Effects on Decongestive Therapy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Effects on GDMT Initiation or Up-titration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Specific Additional Interventions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Renal disease: chronic kidney disease (CKD; eGFR &lt;60 mL/min per 1.73 m<sup>2</sup>) and worsening kidney function (WKF)</b>                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| CKD: Higher doses of loop diuretics are needed in patients with CKD, based on kidney function, <sup>101</sup> and should be followed by early assessment of diuretic response assessment and therapy optimization. <sup>102,103</sup><br>WKF: Decongestion might temporarily lead to moderate increase of creatinine levels, but diuretics should not be stopped if decongestion is effective. <sup>104-106</sup>                                                                                                                 | CKD: GDMT provides benefit in patients with CKD, who are at higher risk of adverse outcomes, though evidence for GDMT is limited for eGFR <30 mL/min per 1.73 m <sup>2</sup> . Continuing BB, ARNI, and SGLT2i in these patients offers clinical benefits without added safety risks. <sup>107,108</sup><br>WKF: Changes in kidney function during GDMT initiation or up-titration are common, but as long as eGFR decreases ≤50%, <sup>109</sup> prognosis is not worsened and long-term eGFR decline is slowed. <sup>110</sup> | CKD: As CKD progresses in HF patients, complications like hyperkalemia may become more common. Combining MRA and SGLT2i and eventually potassium binders may be used to optimize GDMT, depending on eGFR. <sup>111,112</sup> Data on specific dialysis modalities for HF are limited. <sup>113</sup> Ultrafiltration may be considered as bail-out therapy for acute HF to achieve decongestion. <sup>114</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Arrhythmias: atrial fibrillation (AFib) and ventricular arrhythmias (VAs)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| AFib: Effective decongestion may require closer monitoring and therapy adjustments in AFib, because AFib is linked to increased diuretic resistance. <sup>26,115</sup><br>VAs: Decongestion can trigger VAs through hypokalemia and hypomagnesemia. <sup>116</sup> Close electrolyte monitoring and supplementation are advised.                                                                                                                                                                                                  | AFib: Although bBBs do not appear to reduce mortality in AFib patients, GDMT including BBs lowers the incidence and burden of AFib in HF patients. <sup>117,118</sup><br>VAs: Antiarrhythmic drugs (usually amiodarone) used for VAs may complicate GDMT optimization because they affect blood pressure or heart rate. GDMT reduces VAs' risk and burden in HF <sup>116</sup> and should be preferred.                                                                                                                          | AFib: Electrical cardioversion is recommended for hemodynamic instability. <sup>6</sup> In case of lenient rate control in AFib, no adverse effects were found in HF patients, but rhythm control should be considered. <sup>119</sup> Catheter ablation for AFib in HF patients has shown symptom improvement and should be preferred for HF patients with paroxysmal AFib or tachycardia-induced cardiomyopathy, although data on harder outcomes are limited. <sup>120</sup><br>VAs: Electrical cardioversion, intravenous antiarrhythmic drugs, correction of reversible causes, and sedation are initial steps, along with stabilizing hemodynamics and checking for ischemia. <sup>72,116</sup> Antiarrhythmics may help reduce VA burden in ICD patients, and ICD reprogramming with antitachycardia pacing is advised if effective.<br>Catheter ablation is considered for recurrent sustained monomorphic VAs or polymorphic VAs triggered by similar premature ventricular contractions. Secondary-prevention ICDs are indicated for unstable VAs without correctable causes. <sup>6,72</sup> |
| <b>Valvular heart disease (VHD)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Effective decongestion is more difficult in case of VHDs, particularly in AS, MR, TR. <sup>121</sup> Higher loop diuretic doses are often needed. <sup>83</sup> Excessive decongestion in severe AS or mitral stenosis might lead to low cardiac output and hypotension.                                                                                                                                                                                                                                                          | Initiating or up-titrating GDMT is more difficult and with limited prognostic benefit in patients with primary VHDs, often requiring valve intervention first. <sup>83,121</sup> Functional MR may improve with GDMT and should be optimized before considering intervention. <sup>121,122</sup>                                                                                                                                                                                                                                 | AA: Surgical or transcatheter valve replacement is indicated in severe symptomatic AS. Urgent TAVR or balloon valvuloplasty might be required in unstable AHF. <sup>83,121,122</sup><br>MR: For severe secondary MR unresponsive to GDMT, the choice between surgery and TEER depends on risk, MR mechanism, likelihood of repair, and need of concomitant treatment of coronary artery disease or other cardiac conditions. For severe primary MR, surgery is first-line treatment and TEER the alternative for high risk patients. <sup>83,121,122</sup><br>TR: Surgical or transcatheter TR interventions may be proposed to HF patients with symptomatic severe, massive, or torrential TR after optimization of medical therapy. <sup>83,121,122</sup>                                                                                                                                                                                                                                                                                                                                             |
| <b>Hypertension</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Hypertension is often a precipitating factor of congestion and should be promptly treated, particularly in hypertensive pulmonary edema. There are no limitations to diuretic therapy. Intravenous vasodilators may rapidly reduce filling pressures and accelerate decongestion. <sup>31</sup>                                                                                                                                                                                                                                   | GDMT should be prioritized, avoiding drugs with unfavorable hemodynamic or prognostic effects. Early GDMT initiation and rapid up-titration is facilitated by hypertension; vasodilating BBs may be preferred if blood pressure remains >130/80 mm Hg.                                                                                                                                                                                                                                                                           | Non-dihydropyridine calcium-channel blockers (verapamil, diltiazem) should be avoided in HF with reduced ejection fraction owing to their negative inotropic effects. Precipitating factors should be corrected (eg, nonsteroidal antiinflammatory drugs, poor adherence to therapy, uncontrolled pain, obstructive sleep apnea syndrome).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| AHF = acute heart failure; ARNI = angiotensin receptor-neprilysin inhibitor; AS = aortic stenosis; BB = beta-blocker; eGFR = estimated glomerular filtration rate; GDMT = guideline-directed medical therapy; HF = heart failure; ICD = implantable cardioverter-defibrillator; MR = mitral regurgitation; MRA = mineralocorticoid receptor antagonist; SGLT2i = sodium-glucose cotransporter 2 inhibitor; TAVR = transcatheter aortic valve replacement; TEER = transcatheter edge-to-edge repair; TR = tricuspid regurgitation. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**TABLE 2 Evolution in Acute Heart Failure (AHF) Management**

- Disease-modifying oral HF therapies should start during the hospitalization of AHF.
- Patient education and empowerment on appropriate individualized therapies should start before discharge.
- Postdischarge therapy should include up-titration of GDMT, exploring the need of devices and other additional therapies, and implementation of nonpharmacologic treatment, including lifestyle modification.
- Postdischarge monitoring should assess decongestion, symptoms, signs, clinical hemodynamic parameters, vital signs, and biological markers and should include imaging such as echocardiography for cardiac function and structure.
- With optimization of contemporary treatment, depending on hemodynamic parameters, future worsening HF events likely can be managed in an outpatient center.
- Evidence strongly supports that quality of life, hospital readmissions, and survival can be improved with optimized and complete GDMT.
- GDMT shows stronger benefits in AHF than in chronic HF.
- Across the globe, nations must urgently reinforce their HF care infrastructures to ensure better survival of their fellow citizens with HF.

GDMT = guideline-directed medical therapy; HF = heart failure.

stabilized individuals with AHF.<sup>84</sup> In those with significant mitral regurgitation, management with transcatheter edge-to-edge repair (TEER) is an option; however, whenever possible, administration of optimal GDMT is strongly recommended, because reverse cardiac remodeling with GDMT may reduce TEER eligibility by up to 50%.<sup>60,85</sup>

Iron deficiency is a frequent and clinically relevant comorbidity in patients with AHF, affecting up to 80% of patients in the postacute phase, and represents a potentially modifiable condition. Intravenous iron supplementation has been shown to improve functional status and quality of life, and to be associated with a possible reduction in HF readmission and cardiovascular death in patients with HFrEF and HF with mildly reduced ejection fraction with iron deficiency, including when initiated during or shortly after an episode of AHF.<sup>86-89</sup> Detailed recommendations on the management of iron deficiency in HF are provided in a recent comprehensive review.<sup>90</sup>

This discussion is by no means exhaustive; other comorbidities, such as obesity, diabetes, and pulmonary disease, are also highly prevalent, but they are beyond the scope of this review because their impact on the management of AHF itself is less obvious. glucagon-like peptide-1 receptor agonists are not yet recommended during the acute phase of HF hospitalization owing to limited data and potential intolerance; in the postdischarge setting, they may be resumed or initiated in selected patients with obesity—with or without diabetes—to

support cardiometabolic health, although evidence for direct benefit on HF outcomes remains limited.<sup>91-93</sup>

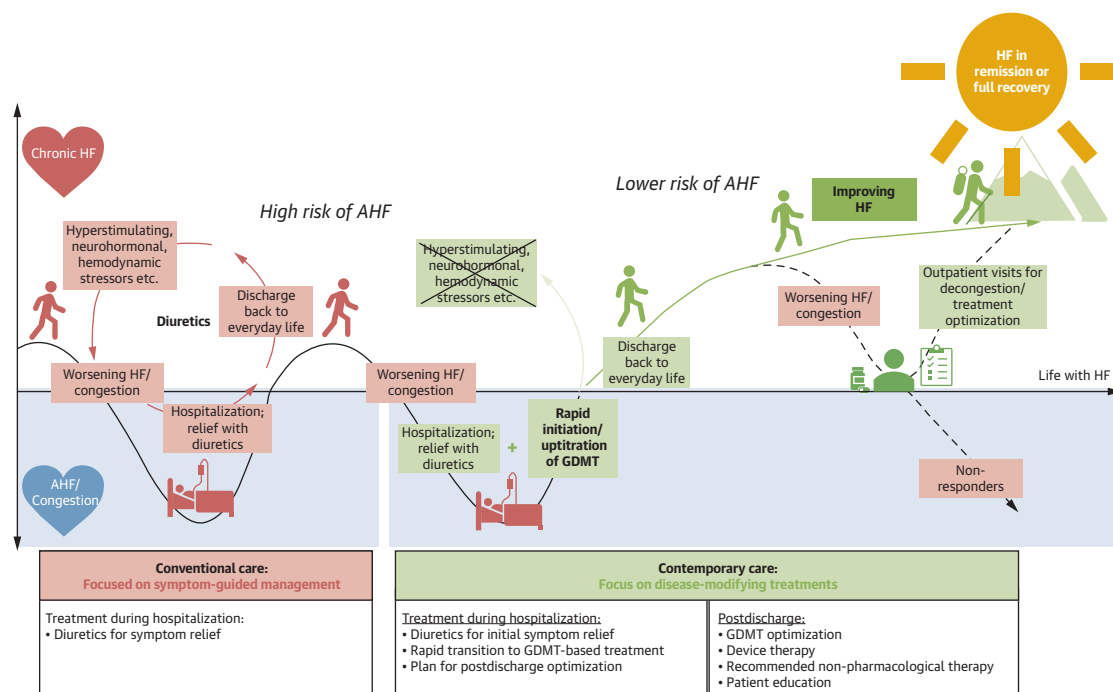
## EVOLVING APPROACHES IN AHF

**Table 2** summarizes the evolution in AHF management, from a conventional diuretic-focused and symptom-guided approach to a contemporary strategy emphasizing disease-modifying therapy with rapid transition to GDMT, targeting lower risk of AHF and, ultimately, disease remission (**Central Illustration**).

## GLOBAL INEQUITIES IN AHF CARE: ONE WORLD, MANY REALITIES

Despite being a leading cause of hospitalization and death worldwide, AHF is still managed within vastly unequal health care landscapes.<sup>2,94-96</sup> This applies both to the initial management in the emergency department or ICU and to post-discharge care, where patients who experienced a life-threatening episode require a comprehensive HF work-up to reduce the risk of avoidable or recurrent AHF admissions. A global heatmap (**Figure 4**) reveals the striking heterogeneity in access to essential diagnostic and therapeutic resources across diverse global regions. No country, regardless of its wealth or infrastructure, offers comprehensive and uniform availability of all recommended tools for AHF and postdischarge care throughout its territory. Although high-income countries demonstrate broader access to essential diagnostics (NPs, echocardiography, chest imaging) and pharmacologic therapies, substantial gaps remain in access to advanced imaging, valvular interventions, palliative care, structured exercise programs, and multidisciplinary HF services. In contrast, many low- and middle-income countries face more foundational challenges—including lack of access to essential medications, limited availability of diagnostic tools, chronic underfunding, and severe shortages of trained personnel, possibly exacerbated by competing public health priorities such as human immunodeficiency virus and tuberculosis. These system-, provider-, and patient-level barriers converge to create fragmented and often inadequate care. Overcoming these bottlenecks requires coordinated context-specific strategies. Health systems must invest in workforce development, equitable distribution of core diagnostics, simplified treatment algorithms, and integration of HF management into broader chronic disease platforms. The heatmap shown in **Figure 4** should catalyze action: It is a moral imperative for global cardiology to ensure that every

## CENTRAL ILLUSTRATION Response to Guideline-Directed Medical Therapy (GDMT) and Long-Term Trajectory



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The majority of patients treated with GDMT respond to therapy, which results in reduction of heart failure (HF) hospitalizations and cardiovascular death and improvement in symptoms, functional capacity, and quality of life (improving HF). These improvements are usually accompanied by reverse remodeling, which may ultimately result in normalization of cardiac function and structure (HF in remission) or—in a few cases—full recovery of HF.<sup>100</sup> However, 20%-30% of patients may not respond favorably to therapies (nonresponders), and experience persisting or worsening symptoms and signs of HF. These subjects display worse outcomes compared with responders even if they do not show active signs of immediate worsening. AHF = acute heart failure.

patient, regardless of geography, has access to timely, evidence-based, and life-saving care.

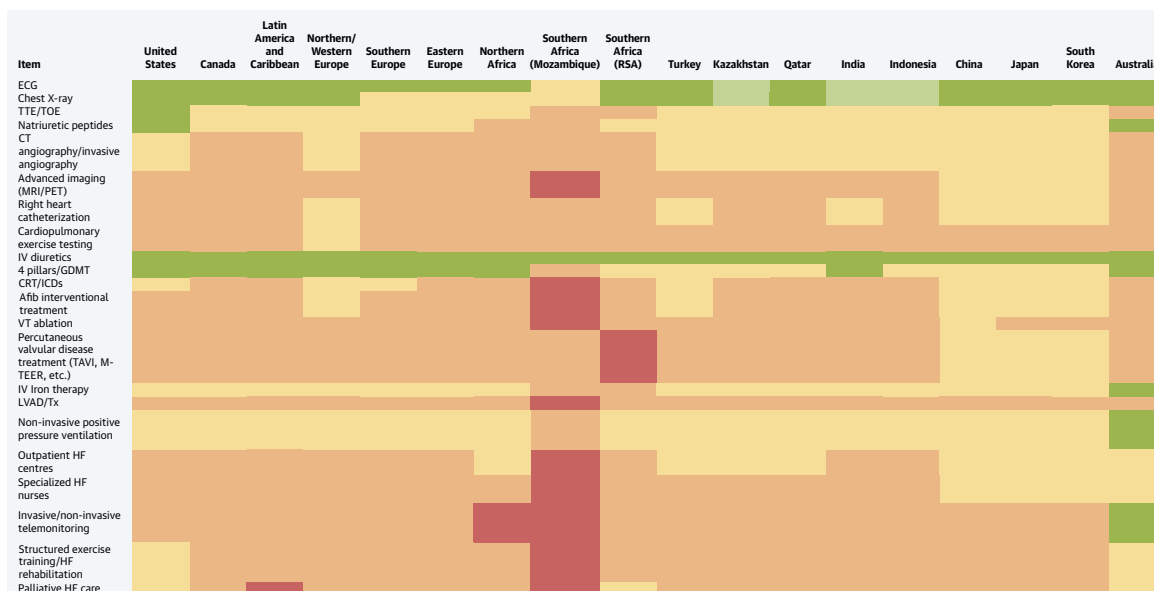
### COMPREHENSIVE CARE OF THE INDIVIDUAL WITH AHF: WHAT MATTERS AT THE FRONT LINE

This global perspective on AHF summarizes the crucial nature of rapid diagnosis, immediate decongestion, and thoughtful application of GDMT. The symptoms of AHF are often frightening for patients and families, and learning of HF diagnosis is similarly anxiety-provoking. A patient-centric approach to AHF care should incorporate team-based approaches that include participation of nursing, hospitalists, and specialists to develop an individualized care plan that not only focuses on the SPLASH acronym (clinical assessment in [Figure 1](#)), but also looks beyond the immediate care to plans for longer-term

management. Hospitalization for AHF represents a key opportunity to optimize long-term outcomes and empower patients for what lies ahead. As noted, a comprehensive discharge plan should include initiation or up-titration of GDMT that is individualized to the patient, a personalized care plan involving all involved (including caregivers), and access to a patient education pack designed to support self-management at home. Clear follow-up with a specialist skilled in the care of HF is essential, not only to titrate medications, but also to assess for eligibility for advanced therapies, including CRTs and ICDs. Beyond medical management, patients benefit greatly from access to peer support groups, where shared experiences foster confidence and hope.

From the patient perspective, improving AHF care is not just about hospital protocols or medication algorithms—it's about building a system that listens, informs, and prepares people to live better with their

**FIGURE 4** Heatmap Showing Global Inequities in Heart Failure Care: One World, Many Realities



The information was derived from national registries, administrative datasets, and expert-driven surveys. The heatmap grades accessibility on a 4-tier system (1: universally available; 2: available in most centers; 3: limited to specialized settings; 4: not available) and provides a synthetic visual guide for regional disparities in HF therapy implementation. Afib = atrial fibrillation; CRT = cardiac resynchronization therapy; CT = computed tomography; ECG = electrocardiography; GDMT = guideline-directed medical therapy; HF = heart failure; ICD = implantable cardioverter-defibrillator; IV = intravenous; LVAD = left ventricular assist device; M-TEER = mitral transcatheter edge-to-edge repair; MRI = magnetic resonance imaging; PET = positron emission tomography; TAVI = transcatheter aortic valve implantation; TOE = transesophageal echocardiography; TTE = transthoracic echocardiography; Tx = transplantation; VT = ventricular tachycardia.

condition. Equitable access, education, and shared decision making must become universal standards to improve the quality and quantity of life for all individuals with AHF.

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